Python INFO-1255 Course Syllabus Spring 2023

"Responsibility for learning belongs to the student, regardless of age" Robert Martin

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Keys to Success: Show Up, Work Hard, Ask for Help

Your Instructor

William A Loring

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E-mail: loringw@wncc.edu (Preferred contact method)

Scottsbluff Office Hours: MW 1-2 pm, M 5-6 pm, TTh 10-11 am or by appointment

Online Office Hours: By appointment: www.calendly.com/loringw

"There are no stupid questions. Ask questions whenever something isn't completely clear. You can't remember what you don't understand."

Tolerate chaos, uncertainty, and vagueness. "Figuring it out" is part of learning.

Class Meeting Times

- **Face to Face Class Location:** Scottsbluff Campus, Room D1
- **Scheduled Online Location:** Zoom link in Blackboard
- **Time:** TTh 12:30-1:45 pm

Prerequisite

None

Catalog Description

This course is an introductory study of computer programming, problem solving methods, and accepted software development practices using Python, an interpreted programming language. Topics include the fundamentals of Python procedural programming, object-oriented programming, and introduction of some advanced features of Python. This course prepares the student for further study in Computer Science, CyberSecurity, IoT, GIS Software, Robotics, and Engineering.

3 semester hours

(3/45/0/0/0/0) See Figure 1

Course Objectives

Using this course as an instructional medium, the instructor will:

1. Define and explain common Python syntax, terms and concepts.
2. Explain and demonstrate how to design, develop and execute Python programs.
3. Demonstrate and model how to solve programming problems based on case studies.
4. Model self-directed and lifelong learning.

Student Learning Outcomes

Upon completion of this course, the student will be able to:

1. Recognize and define common Python syntax, terms, and concepts. [GE 1, 2; INFO 3]
2. Create and compile example Python programs. [GE 2; INFO 3]
3. Demonstrate the concepts of object-oriented programming, problem solving, and debugging by creating functional Python applications based on case studies. [GE 2; INFO 3]
4. Self-direct their learning while gaining an ongoing interest in learning more about programming. [GE 5; INFO 3]

Instructional Materials

All instructional materials, readings, and tutorials are in the Blackboard course. There is not a book required for this class.

OER (Open Educational Resources) are freely accessible, openly licensed text, media, and other digital assets that are useful for teaching, learning, and assessing as well as for research purposes. This is a concept that allows for open source sharing of educational materials for no cost.

There is a wealth of free educational material available for Python.

Some of the materials used in this course are adapted from OER (Open Educational Resources) resources. The OER materials used in this course are adapted under Creative Commons license https://creativecommons.org/licenses/by-nc-sa/3.0/us.

The following books were major sources for this class. You are not required to read these books; the links to the books are provided as a resource if you wish to read further.

- The Coder's Apprentice, Creative Commons, Pieter Spronck
- Python for Everybody, Creative Commons, Dr Charles Severance
- A Practical Introduction to Python Programming, Creative Commons, Brian Heinold
- ISTE 1359 Introduction to Scripting Languages: Python, Creative Commons, Richard Baldwin

Course Schedule

Course content and schedule may change.

Week	Learning Activities	Assignments
Week 1 01/16 - 01/22	Introduction Discussion Introduction to Course Chapter 1: Introduction to Programming Chapter 2: Getting Started with Python TTh 2:00-3:15 pm: Face to Face or Live Stream	Getting Started Activities Professional Communication Getting Started Quiz Install Python Chapter 2 Getting Started with Python Tutorials Chapter 2 Math
Week 2 01/23 - 01/29	Chapter 2: Getting Started with Python Chapter 3: Decisions TTh 2:00-3:15 pm: Face to Face or Live Stream	Pseudocode Tutorial Pseudocode Quiz Chapter 2 MPG Chapter 2 Leave a Tip Chapter 3 Decisions Tutorials Chapter 3 Positive or Negative

		Chapter 2 Quiz
Week 3 01/30 - 02/05	Chapter 3: Decisions TTh 2:00-3:15 pm: Face to Face or Live Stream Exit Ticket Tutorial	Chapter 3 Payroll Advice with Overtime Chapter 3 Purchase Price Chapter 3 Temperature Converter Chapter 3 BMI Calculator Chapter 3 Quiz
Week 4 02/06 - 02/12	Chapter 4: Loops TTh 2:00-3:15 pm: Face to Face or Live Stream	Chapter 4 Loops Tutorials Chapter 4 Paul's Purchasing Pandemonium Chapter 4 Irene the Insect Collector
Week 5 02/13 - 02/19	Chapter 4: Loops TTh 2:00-3:15 pm: Face to Face or Live Stream Exit Ticket Tutorial	Chapter 4 Live Stream Service Chapter 4 Pennies for Penny Meow Facts API Tutorial with Python Chapter 4 Quiz
Week 6 02/20 - 02/26	Chapter 5: Functions TTh 2:00-3:15 pm: Face to Face or Live Stream	Chapter 4 Rock, Paper, Scissors, Lizard, Spock Chapter 5 Function Tutorials Chapter 5 Joke and a Punch Line Chapter 5 Dave's Dice Game
Week 7 02/27 - 03/05	Chapter 5: Functions TTh 2:00-3:15 pm: Face to Face or Live Stream Exit Ticket Tutorial	Chapter 5 Text Adventure Chapter 5 Carling 's Cost Calculator Chapter 5 Peggy's Paint Job Estimator Chapter 5 Quiz

Week 8 03/06 - 03/10	Chapter 6: Lists, Tuples, and Dictionaries TTh 2:00-3:15 pm: Face to Face or Live Stream	Chapter 6 Lists, Tuples, and Dictionaries Tutorials Chapter 6 Modify List Chapter 6 Lists Exercise Chapter 6 Speedtest Guild Project Team Process Skype and Zoom Lab
03-11 – 03-19		
Week 10 03/20 – 03/26	Chapter 6: Lists, Tuples, and Dictionaries TTh 2:00-3:15 pm: Face to Face or Live Stream Exit Ticket Tutorial	Chapter 6 Dave's Dice Game Reloaded Chapter 6 Pete's Pizza Palace Chapter 6 English to Spanish Dictionary Guild Project Kat's Lemonade Stand Part 1 Chapter 6 Quiz
Week 11 03/27 – 04/02	Chapter 7: Object-Oriented Programming TTh 2:00-3:15 pm: Face to Face or Live Stream	Chapter 7 OOP Tutorials Chapter 7 Car Class Chapter 7 Network Scanner OpenWeatherMap Tutorial Guild Project Kat's Lemonade Stand Part 2
Week 12 04/03 – 04-09	Chapter 7: Object-Oriented Programming TTh 2:00-3:15 pm: Face to Face or Live Stream Exit Ticket Tutorial	Chapter 7 Product Class Chapter 7 Chat Client and Server Chapter 7 Quiz Guild Project Kat's Lemonade Stand Part 3

Week 13 04/10 – 04/16	Chapter 8: GUI TTh 2:00-3:15 pm: Face to Face or Live Stream	Chapter 8 GUI Tutorials Chapter 8 Tkinter Web API OpenWeatherMap API Guild Project Kat's Lemonade Stand Part 4
Week 14 04/17 – 04/23	Chapter 8: GUI TTh 2:00-3:15 pm: Face to Face or Live Stream IT Jobs Discussion	OpenWeatherMap API GUI Tutorial Chapter 8 Quiz Guild Project Kat's Lemonade Stand Part 5
Week 15 04/24 – 04/30	Chapter 9: Files TTh 2:00-3:15 pm: Face to Face or Live Stream Lessons Learned Discussion	Chapter 9 Files Tutorials Chapter 9 Guest Book Chapter 9 CSV to Joke Program Tutorial Chapter 9 Quiz Guild Project Kat's Lemonade Stand Part 6
Week 16 05/01 – 05/05		Music Player Guild Project Kat's Lemonade Stand Part 7
Finals 05/08 - 05/12	Finals Week	Final Project: Guild Project Kat's Lemonade Stand

Academic Integrity

The academic integrity policy for this course includes the Institutional Academic Integrity Policy listed at the end of this document.

1. Do your own work.

2. You can ask for help if you get stuck. It is OK to have a study buddy to help with problems or issues. It is not OK to turn in the same assignment as someone else.
3. If you use someone else's work for a small quote or reference, cite the source.
4. Use your own words.
5. Do your own work. We are here to learn. You can't learn without doing the work.

Minor Violations: First offense: Grade of 0 for the assignment.

Major Violations: Second offense: Grade of F for the class.

Do your own work.

Institutional Credit Hours and Out-of-Classroom Preparation

WNCC credit hour assignments comply with the Federal Credit Hour definition.

At WNCC, a semester **credit hour** is normally granted for the satisfactory completion of one 50-minute session (contact hour) of classroom instruction per week for a semester of not less than 15 weeks; two to three contact hours are required for laboratories to award a single credit hour. Other differences may occur depending on instructional delivery.

On average, for lecture courses, one credit hour requires three hours of effort on the part of the student per week – **one hour of class and two hours of out-of-class time to accomplish readings and out-of-class assignments in preparation for successful completion of the course requirements.** Some students may require more preparation time for some classes and for some classes, less time.

For a three-hour lecture course, a student should plan on a minimum of three hours a week in class and a minimum of six hours a week spent on out-of-class preparation.

Institutional Academic Integrity

Academic integrity forms a fundamental bond of trust between colleagues, peers, teachers, and students, and it underlies all genuine learning. At WNCC, there is no tolerance for plagiarism or academic dishonesty in **any** form, including unacknowledged "borrowing" of proprietary material, copying answers or papers, using crib sheets, unauthorized help during exams, altering tests, or passing off someone else's work as one's own.

A breach of ethics or act of dishonesty can result in:

- failure of a paper or an exam within a course (instructor level)

- failure of an entire course (institutional level)
- suspension or expulsion from the college (institutional level)

Any form of academic dishonesty represents a grave breach of personal integrity and of the rules governing WNCC's community of learners. **Academic dishonesty includes, but is not limited to**:

- cheating in any form
- plagiarizing in any form
- aiding someone else in cheating or plagiarizing

Equal Access

Western Nebraska Community College seeks to make all programs and services, including electronic and information technology, accessible to people with disabilities. In this spirit, and in accordance with the provisions of Sections 504 and 508 of the Rehabilitation Act and the Americans with Disabilities Act (ADA), WNCC provides students, faculty, staff, and visitors with reasonable accommodations to ensure equal access to the programs and activities of the college. For assistance or further information, students with disabilities should contact the Counseling Director at (308) 635-6090. Helpful information is also available in the *"Transition Guide for Students with Disabilities"* on the WNCC Web site under Disability Services.

Title IX Statement

Title IX of the Education Amendments of 1972 prohibits discrimination on the basis of sex in any educational institution that receives federal funding. WNCC strictly prohibits any form of sexual harassment, which includes sexual harassment, sexual assault, dating violence, domestic violence, and stalking. All reported incidents will be thoroughly investigated and those found responsible dealt with as necessary, whether criminally charged or handled through the College's sexual harassment grievance and investigation procedure. College policy also prohibits retaliation against a person for reporting sexual harassment or participating in the investigation or resolution of such a complaint. Help and support are available for any student who experiences any form of sexual harassment.

Questions or concerns about a student's rights, the resources available, or to file a Title IX complaint, should contact the College's Title IX Coordinator at 308.635.6105. More information about reporting options and resources at WNCC and the community are available at www.wncc.edu/about-wncc/consumer-information.

If a student wishes to speak to someone confidentially, they should contact the Counseling Director at the WNCC Counseling Center on the Scottsbluff campus or by calling 308.635.6090. Appointments are available at all WNCC locations.

Discrimination, Harassment, and Retaliation

WNCC is committed to providing a college environment free from harassment, discrimination, and retaliation, and all students, employees, volunteers, and visitors are prohibited from participating in any harassment, discrimination, or retaliation based on any protected class status.

If a student feels that he or she is the victim of discrimination, harassment, or retaliation, it is important to remember that there are supportive people at WNCC who are resources for discussing and helping to clarify what constitutes discrimination, harassment, or retaliation and the action steps that can be taken. The College has established both formal and informal procedures to report complaints.

Students should not wait to report conduct of concern until harassment becomes sufficiently serious (i.e., severe, pervasive, or persistent) to create a hostile environment. Off-campus harassment, misconduct, or violence that creates a hostile environment on campus should be brought to the attention of the College.

To file a complaint or get help, students should seek assistance through WNCC's Institutional Civil Rights Officer and Title IX Coordinator (308-635-6105). More complete information about the reporting process can be found in the WNCC Student Handbook.

Student Code of Conduct

As an educational institution of higher education, WNCC seeks to provide the best educational environment for its students, faculty, and staff. It is mindful that academic freedom is an essential element of college life. The purpose of the Student Code of Conduct is to help ensure a safe and educationally productive environment for students, employees, and visitors. Conduct that adversely affects a student's responsible membership in the academic community shall result in appropriate disciplinary action.

Please see the *Student Handbook (www.wncc.edu/student-life/student-handbook)* for a complete copy of the WNCC Student Code of Conduct.

Basic Needs Security

Students who believe their academic performance is being negatively impacted due to difficulty accessing sufficient food or finding a safe and stable place to live are encouraged to contact the Dean of Students' office at 308-635-6123. This office will put students in contact with community partners who may be able to aid in these areas.

Audio/Visual Recordings

Except where a student is entitled to make an audio or video recording of class lectures and discussions as an educational accommodation determined through the student's interactive process with college disability services, a student may not record lectures or classroom discussions unless written permission from the class instructor has been obtained and all students in the class as well as guest speakers have been informed that audio/video recording may occur.

A student granted permission to record may use the recording only for his or her own study and may not publish or post the recording on YouTube or any other medium or venue without the instructor's explicit written authorization.
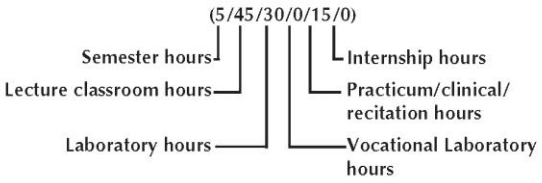
Figure 1: Explanation of Credit Diagram